Quiz 2

Student Name: ______________________ **Points:** 30

A Number: ______________________ **Duration:** 20 Minutes

1. Choose what is true about `nn.Module`
   (a) `nn.Module` is a function used in PyTorch to train a neural network.
   (b) `nn.Module` is an abstract class used to implement a Neural Network Layer.
   (c) `nn.Module` is a data type for Neural Networks in PyTorch
   (d) `nn.Module` is a neural network parameter.
2. Neural network weights in PyTorch can be of type:
   (a) `nn.Parameter`
   (b) `nn.Module`
   (c) `nn.Conv2D`
   (d) `nn.Weight`
3. Hooks in PyTorch are
   (a) functions
   (b) variables
   (c) tensors
   (d) classes
4. What is the difference between parameters and buffers in the realm of PyTorch `nn.Module`?

Parameters and buffers are both Tensors and member variable of Neural Network class `nn.Module`. However parameters are learable tensors and requires gradient to be true by default, i.e. `requires_grad=True`. That is not true for buffers. However, buffers are variables that do store some important information (for example, scaling parameters) that you would need to reload when you when load a trained model for making an inference. Hence, in such a case you need to register both: parameters and buffers in class definition. But buffer neglects operations to compute gradients and update its values. Only parameters are involved in gradient computations.

5. What happens if you don't register a member variable tensor as parameters in PyTorch `nn.Module`?

> If we do not register a member variable tensor as parameters in PyTorch `nn.Module` it doesn't become learnable and won't be updated via backpropagation when training of the model is done.

6. When are forward hooks executed?

   (a) The hook are called every time after `backward()`.

   (b) The hook are called just before calling `forward()`.

   (c) The hook are called every time after weights are updated..

   (d) The hook are called every time after `forward()`.

7. Is it possible to implement a neural network in PyToch without inhering from `nn.Module`?

   (a) Yes

   (b) No

8. Consider a Network definition

```python
class Network(nn.Module):
        def __init__(self):
                super().__init__()
                self.layer1 = nn.Linear(10, 20)
                self.layer2 = nn.Linear(20, 5)
        def forward(self, x):
                return self.layer2(self.layer1(x))
```

   How many trainable parameters are there in the above network (including biases)?

   (a) 325

   (b) 300

   (c) 55

   (d) 400

9. Imagine a dataset consisting of PNG images with four channels: RGBA. You decided to use a fully connected neural network for creating an image classfier. There 10000 images in your training dataset. Each image has dimension 200×200. How many features will be there in the input feature matrix to the neural network?

   (a) 10000

   (b) 1600000000

(c) 40000

(d) 400000000

10. Minpooling downsamples the image data.

    (a) True

    (b) False

11. An example of order-5 Tensor:

    A batch of colored movies.(batch size, frames, height, width, channels)
    or
    A single 3D colored movie. (frames, height, width, depth channels)

12. Consider a convolution layer definition

```
conv = nn.Conv2d(in_channels=4,
                 out_channels=6,
                 kernel_size=3,
                 stride=2,
                 padding=1)
```

    What is the dimension of an example tensor that can serve as an input to the above convolution layer?

    (a) `torch.Size([2, 1, 5, 6])]`

    (b) `torch.Size([2, 6, 32, 32])]`

    (c) `torch.Size([2, 3, 5, 6])]`

    (d) `torch.Size([2, 4, 32, 32])]`

13. For the convolution operation in Q12, how many filters will be learnt upon training?

    (a) 4

    (b) 3

    (c) 6

    (d) 2

14. If you are performing regression with a neural network, what should be the final layer in a neural network?

    (a) A linear layer

    (b) A sigmoid layer

(c) A relu layer

(d) A tanh layer

15. Applying max pooling with a kernel size of 3×3 to the image of size 5×5 with a stride of 2, the resulting output has size

(a) 2×2

(b) 3×3

(c) 5×5

(d) 3×5